

IP Addressing

IPv4, IPv6, Subnetting and CIDR Fully Explained

1. Why IP Addressing is everything.

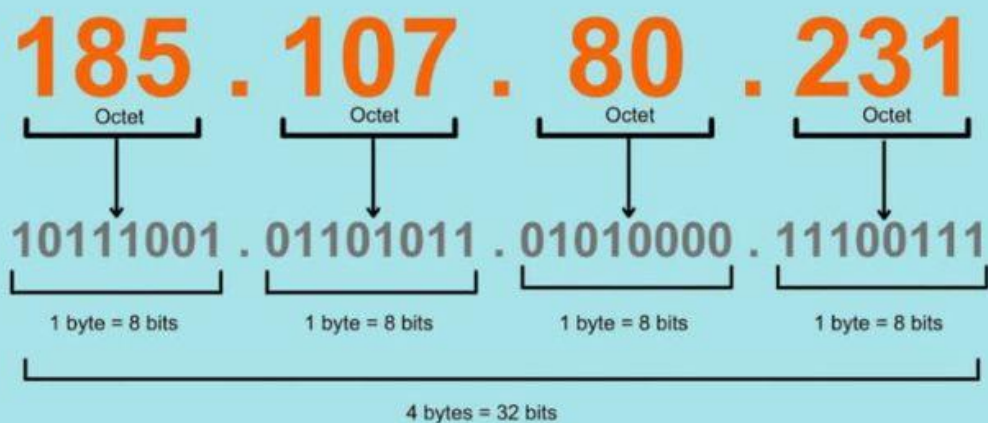
Every device that connects to a network needs an address. Without an address there is no way to identify where data should go or where it came from. IP addressing is the system that gives every networked device a unique identity and understanding it is not optional for anyone who wants to work in cloud infrastructure, Azure administration, or any networking role.

When you configure an Azure virtual network, assign an IP to a virtual machine, write a firewall rule, or troubleshoot why two servers cannot reach each other, you are working with IP addressing every single time. This document covers the complete theory from how an IPv4 address is structured, through subnetting and CIDR notation, all the way to IPv6 and how these concepts apply in Azure environments.

Introduction to IP Address !

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An IP address (Internet Protocol address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.



It's like a digital street address that allows devices to send and receive data over the internet or a local network

There are two versions of the Internet Protocol in use today. IPv4 has been the dominant version since the 1980s. IPv6 was developed to solve the address exhaustion problem that IPv4 faced as the internet grew far beyond what its designers anticipated.

2. IPv4: Structure and Addressing

How an IPv4 address is built

An IPv4 address is 32 bits long, written as four groups of decimal numbers separated by full stops, for example 192.168.10.25. Each group is called an octet because it represents eight binary bits. An octet can hold a value from 0 to 255, because eight bits can represent 256 unique values (from 00000000 to 11111111).

Octet 1 (Decimal)	Octet 2 (Decimal)	Octet 3 (Decimal)	Octet 4 (Decimal)
192	168	10	25
11000000	10101000	00001010	00011001

Every IP address has two parts: the network portion, which identifies which network the device belongs to, and the host portion, which identifies the specific device within that network. The subnet mask tells the device where the dividing line is.

IPv4 Address Classes

Before CIDR notation arrived, IP addresses were divided into five classes based on the value of the first octet. Understanding classes helps you recognise address ranges quickly and understand how the internet's addressing history evolved.

Class	First Octet Range	Default Subnet Mask	Max Networks	Max Hosts Per Network
A	1 to 126	255.0.0.0 (/8)	126	16,777,214
B	128 to 191	255.255.0.0 (/16)	16,384	65,534
C	192 to 223	255.255.255.0 (/24)	2,097,152	254
D	224 to 239	Not applicable	N/A	N/A — multicast groups only
E	240 to 255	Not applicable	N/A	N/A — reserved for research

Class A addresses provide a small number of networks but a huge number of hosts per network, making them suited for very large organisations. Class C addresses provide many networks but only 254 hosts each, making them suited for small organisations. The class system was replaced by CIDR in the 1990s because it wasted enormous amounts of address space.

The address 127.0.0.1 is missing from the Class A range above because it is permanently reserved as the loopback address. Pinging 127.0.0.1 tests your own network stack without sending any traffic onto the actual network.

Private IP Addresses Ranges (RFC 1918)

Private IP addresses are ranges reserved for use inside private networks. They are not routable on the public internet, meaning a router connected to the internet will not forward packets addressed to these ranges. This is intentional and defined in RFC 1918.

Class	Private Range	Subnet Mask	Common Use
A	10.0.0.0 to 10.255.255.255	255.0.0.0 (/8)	Large enterprise networks and Azure VNets
B	172.16.0.0 to 172.31.255.255	255.240.0.0 (/12)	Medium businesses and ISP networks
C	192.168.0.0 to 192.168.255.255	255.255.0.0 (/16)	Home networks and small offices

When a device with a private IP address needs to reach the internet, NAT (Network Address Translation) is used. A router or firewall replaces the private source IP with its own public IP, keeps a record in a translation table, and reverses the process when the response comes back. This is how your home router allows ten devices to share a single public IP address.

In Azure, all virtual network address spaces use private IP ranges. Communication inside a VNet uses private IPs. Access to the internet is provided through Azure NAT Gateway or a public IP assigned directly to a resource.

3. Subnets Masks and Subnetting

What a subnet mask does

A subnet mask is a 32-bit number that tells a device which portion of an IP address identifies the network, and which portion identifies the host. The mask uses binary 1s for the network bits and binary 0s for the host bits.

For the address 192.168.1.50 with subnet mask 255.255.255.0, the first three octets (192.168.1) are the network portion and the last octet (50) is the host portion. In binary, 255.255.255.0 is: 11111111.11111111.11111111.00000000.

When your device wants to send data to another IP address, it applies the subnet mask to both addresses. If the network portions match, the destination is on the same local network and data is delivered directly using MAC addresses at Layer 2. If they do not match, the data is sent to the default gateway (the router) for Layer 3 routing.

CIDR Notation

CIDR (Classless Inter-Domain Routing), defined in RFC 4632, replaced the rigid class-based system and gave network designers precise control over address allocation. Instead of being locked into a Class A, B, or C mask, you can now define exactly how many bits belong to the network.

CIDR notation appends the number of network bits to the IP address after a forward slash. The address 192.168.1.0/24 means the first 24 bits are the network portion. This leaves 8 bits for hosts. Eight bits gives 256 total addresses, minus 2 reserved (the network address and broadcast address), leaving 254 usable host addresses.

CIDR	Subnet Mask	Network (Example)	Usable Hosts	Broadcast (Example)
/8	255.0.0.0	10.0.0.0	16,777,214	10.255.255.255
/16	255.255.0.0	192.168.0.0	65,534	192.168.255.255
/24	255.255.255.0	192.168.1.0	254	192.168.1.255
/25	255.255.255.128	192.168.1.0	126	192.168.1.127
/26	255.255.255.192	192.168.1.0	62	192.168.1.63
/27	255.255.255.224	192.168.1.0	30	192.168.1.31
/28	255.255.255.240	192.168.1.0	14	192.168.1.15
/29	255.255.255.248	192.168.1.0	6	192.168.1.7
/30	255.255.255.252	192.168.1.0	2	192.168.1.3

A Practical Subnetting Example

A company is given the network 192.168.10.0/24 and needs to divide it into four equal subnets. Here is the calculation:

1. You need 4 subnets. You borrow 2 bits from the host portion because 2 to the power of 2 equals 4.
2. The new prefix becomes /26 (24 original network bits plus 2 borrowed bits).

3. Each /26 subnet has 6 host bits, giving 64 total addresses and 62 usable host addresses.

Subnet	Network Address	Usable Host Range	Broadcast	Prefix	Hosts
1	192.168.10.0	192.168.10.1 to 192.168.10.62	192.168.10.63	/26	62
2	192.168.10.64	192.168.10.65 to 192.168.10.126	192.168.10.127	/26	62
3	192.168.10.128	192.168.10.129 to 192.168.10.190	192.168.10.191	/26	62
4	192.168.10.192	192.168.10.193 to 192.168.10.254	192.168.10.255	/26	62

In Azure, when you create a VNet and define subnets, this calculation happens every time. Remember: Azure also reserves 5 addresses per subnet for its own use, so a /26 in Azure gives you 62 minus 5 = 57 usable addresses for your virtual machines.

4. IPv6: Why It Exists and How It Works

The Problem IPv6 solves

IPv4 can support approximately 4.3 billion unique addresses. When the Internet Protocol was designed in the early 1980s, that seemed like more than enough. It was not. By the early 2000s, explosive internet growth meant the pool was running dry. IANA allocated the last unassigned IPv4 address blocks to regional registries in February 2011.

IPv6 was designed to solve this permanently. With 128-bit addresses, it can produce 340 undecillion unique addresses. That number has 39 digits. It is enough to assign multiple IP addresses to every atom on the surface of the Earth.

IPv6 Address Format

An IPv6 address is 128 bits written as eight groups of four hexadecimal digits separated by colons. For example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334.

Two shortening rules make addresses easier to write and read:

- Leading zeros inside each group can be dropped. So 0db8 becomes db8 and 0000 becomes 0.

- One or more consecutive groups of all zeros can be replaced with a double colon (::), but only once per address. So, 2001:0db8:85a3:0000:0000:8a2e:0370:7334 becomes 2001:db8:85a3::8a2e:370:7334.

IPv6 address types:

- Unicast: Identifies a single interface. Traffic goes to exactly one device.
- Multicast: Identifies a group of interfaces. Traffic goes to every member of the group. IPv6 uses multicast where IPv4 used broadcast.
- Anycast: Identifies a group of interfaces. Traffic goes to the nearest one in routing terms. Used for load distribution and redundancy.

There is no broadcast in IPv6. Broadcast is replaced entirely by multicast, which reduces unnecessary traffic on large networks.

5. IPv4 vs IPv6 Comparison

IPv4	VS	IPv6
32 bits long address		128 bits long address
Total 2^{32} addresses		Total 2^{128} addresses
Manual & DHCP Configuration		Auto IP Configuration
20-60 bytes variable header		40 bytes fixed header
No end to end integrity		End to end connection integrity
No specific security mechanism		Uses IPSec
Uses Checksum		No Checksum
Uses IP Classes and VLSM		No IP Classes and VLSM
No packet identification (QoS)		IPv6 QoS
Fragmentation by sender and forwarding router		Fragmentation by only sender

6. Network Address Translation (NAT)

NAT allows multiple devices with private IP addresses to share one public IP address when accessing the internet. It is one of the main reasons the internet did not completely collapse under IPv4 address exhaustion.

In a typical home network, your router has one public IP address from your ISP. Every device inside has a private address like 192.168.1.x. When your laptop sends traffic to a website, the router replaces the private source IP with its own public IP, records the private IP and port number in a NAT table, and forwards the packet. When the website responds, the router looks up the NAT table, swaps the destination back to your laptop's private IP, and delivers it.

NAT Type	Also Known As	When It Is Used
Static NAT	One-to-one NAT	Maps one private IP permanently to one public IP. Used for servers that must always be reachable at the same public address.
Dynamic NAT	Many-to-many NAT	Maps private IPs to a pool of public IPs. Used by enterprises with multiple public IPs.
PAT / Overload	Many-to-one NAT	Maps many private IPs to a single public IP using different port numbers. Used in home routers and Azure NAT Gateway.

PAT (Port Address Translation), also called NAT Overload, is the form used in home routers and Azure NAT Gateway. It uses different source port numbers to track multiple simultaneous sessions through a single public IP address.

7. IP Addressing in Azure

Azure Concept	IP Type	What It Does
Virtual Network (VNet)	Private (RFC 1918)	Defines the address space for your Azure network. You choose a private CIDR range.
Subnet	Private (RFC 1918)	Divides the VNet. Azure reserves 5 IPs per subnet for internal use.
VM NIC — Private IP	Private	The internal IP of a virtual machine, used for communication inside the VNet.
VM NIC — Public IP	Public	An optional public IP for internet-facing access. Can be static or dynamic.
Azure Load Balancer	Public or Private	Frontend IP that receives traffic and distributes it to backend VMs.
Azure NAT Gateway	Public	Provides outbound internet access for private VMs through SNAT.
Azure Bastion	Private and Public	Secure RDP and SSH to VMs without exposing individual public IPs.